BetterGov.PH

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HOW TO BUILD A NATIONAL CHATBOT · ENGINEERING NOTES

Route first. Spend tokens wisely.

Jev screens requests, narrows the tools and filters returned context.
The answer AI focuses on retrieval and explanation.

Built by 1 person.
Zero funding.

OBSERVED PSA ROUTE · TOOL DEFINITIONS

52 → 7 tools

87% less tool-definition text

Full catalog: 32,950 characters
Selected catalog: 4,193 characters

INPUT SCREENING

17 → 1

Judgments → Jev request

Scope, spam, injection, source relevance and lookup type.

REQUESTS BLOCKED BY SCREENING

0

Answer-model calls

Also zero source lookups.
The app returns a friendly redirect.

Measured catalog size, not a billing claim. 87% is the reduction in serialized tool-definition characters for one recorded route. Actual tokens and cost depend on the tokenizer, conversation, evidence and retries.

Two Jev checkpoints. One bounded research loop.

Jev judgment · AI generation · Source systems

01 · BROWSER / PWA → APP

Ask → validate request & capacity

English, Filipino or Taglish. Code checks the request boundary.

Code owns the decisions.
Jev scores. AI plans. Adapters execute.

02 · JEV / INPUT GATE

Screen the request. Recommend sources.

Typed judgments cover scope, spam and bypass attempts.
Code applies thresholds and selects relevant source families.

block

Blocked → friendly redirect

Stop before tool selection or answer generation.
Jev screening still has its own cost.

03 · TOOL CATALOG → AI PLANNER

Choose a permitted tool and arguments

The AI sees the selected catalog, then requests an exact lookup.

04 · APPLICATION CODE

Validate & execute a read-only lookup

Adapters enforce argument schemas, limits and allowed tools.

lookup

Existing knowledge projects

Built and maintained independently over months.

- Documents + public datasets
- Source-owned retrieval — RAG / semantic search / tables / exact records
- Published APIs + MCP tools

05 · JEV / CONTEXT GATE

Screen the returned records and metadata

Relevance, prompt injection and contradiction checks run in a batch. Code keeps useful context and flags conflicts.

KEEP · KEEP + FLAG CONFLICT · DROP

results

Documentation + llms.txt

Integration-time guidance for the configured catalog and adapters.
Live answers come from API/MCP results.

MORE DETAIL NEEDED / RESULT DROPPED → BOUNDED LOOKUP LOOP

06 · ACCEPTED CONTEXT + CITATIONS → AI

Consolidate multiple sources into one answer

Combine the accepted information, preserve limitations and request another lookup when more detail is needed.

WHEN RESEARCH CANNOT ANSWER

No accepted results at completion?

Return a friendly reply to rephrase or try again.
Unsupported factual drafts are withheld.

Bounded calls + duplicate-call suppression.
Filtered text is excluded from AI context.
A different permitted lookup may still help.

07 · APP → RESIDENT

Stream the answer, references & optional Debug

Source pages open directly; the main website is the fallback.

Routing normally exposes up to 4 of 10 source families. An uncertain route can expose all 10.
Rejected results return a short tool error to the AI; retries remain subject to the app's limits.

Retrieval stays with the systems that built the knowledge.

10 source families · 9 domains

juris.ph
Laws & court decisions

bills.juris.ph
Bills & authors

budget.bettergov.ph
National budgets

statistics.bettergov.ph
PSA datasets + classification codes

officials.bettergov.ph
Officials & election records

bettergov.ph
Government services

api.dpwh.bettergov.ph
Public works API

flood-control.bettergov.ph
Flood control projects

asean.bettergov.ph
Regional indicators

Months of collection, cleaning and indexing are reused. **No new central vector database.**

Where the cost and token efficiency comes from

Stop avoidable work
Blocked requests never reach the answer model or knowledge-source APIs.

Send a smaller tool catalog
Relevant tool schemas replace the full catalog in routed model calls.

Keep answer context focused
Jev rejects whole results. The app also bounds and abbreviates source text.

Net benefit = avoided AI work − Jev overhead

Jev adds screening requests. Measure total cost and latency per completed answer; this example is not an end-to-end savings benchmark.